

NEUROLOGY & MEMORY DISORDERS CLINIC

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NEUROLOGY CONSULTATION — COGNITIVE ASSESSMENT

Patient Name: Ester635 Echevarría842
Date of Birth: June 20, 1927 (Age: 92 years)
Sex: Female
Medical Record #: MRN-847293
Date of Consultation: September 18, 2019
Referring Physician: Dr. Margaret R. Thompson, Internal Medicine
Consultation Requested Primary Care / Family
By:

CHIEF COMPLAINT

Progressive memory loss and confusion.

HISTORY OF PRESENT ILLNESS

Ms. Echevarría is a 92-year-old Spanish-speaking female who presents to the Memory Disorders Clinic accompanied by her daughter and granddaughter for evaluation of progressive cognitive decline over the past 18-24 months. The family reports that the patient has been experiencing increasing forgetfulness, particularly with short-term memory. She frequently misplaces items, repeats questions within the same conversation, and has difficulty remembering recent events or appointments.

Of particular concern, the family notes that Ms. Echevarría has become disoriented in previously familiar settings. Approximately three months ago, she became lost while walking in her own neighborhood, an area she has lived in for over 30 years. A neighbor found her and brought her home. More recently, she has had difficulty recognizing familiar places when being driven to appointments.

The family also reports confusion regarding medication management. Despite using a pill organizer, the patient has been found taking medications at incorrect times or missing doses altogether. On at least two occasions in the past month, family members discovered that she had taken her morning medications twice. Given her complex medication regimen including warfarin for atrial fibrillation and insulin for diabetes, this has become a significant safety concern.

The patient demonstrates preserved remote memory and can recall events from her youth and middle adulthood with relative clarity. However, she has difficulty with executive functions such as planning meals, managing finances (which she can no longer do independently), and following multi-step instructions. The family reports that her personality remains largely intact, though she has become more passive and less engaged in previously enjoyed activities such as cooking and watching telenovelas.

There have been no reports of visual or auditory hallucinations, no significant behavioral disturbances, and no symptoms suggestive of depression. Sleep-wake cycle is generally preserved. There is no history of head trauma, stroke symptoms, or seizures. Family history is notable for dementia in the patient's mother, who developed symptoms in her late 70s.

PAST MEDICAL HISTORY

- Alzheimer's disease (diagnosed approximately 12 months ago)
- Atrial fibrillation (on anticoagulation)
- Type 2 diabetes mellitus with complications including diabetic retinopathy and diabetic renal disease
- Chronic kidney disease, stage 1
- Anemia (on erythropoietin therapy)
- Hypertriglyceridemia and hyperlipidemia
- Metabolic syndrome
- Osteoarthritis of the knees
- Chronic sinusitis
- History of recurrent rectal polyps and colon polyps
- Acute bronchitis (recent)
- Viral sinusitis (current)

SURGICAL HISTORY

- Electrical cardioversion for atrial fibrillation
- Catheter ablation of cardiac tissue
- Multiple colonoscopic polypectomies

CURRENT MEDICATIONS

- Galantamine 4 mg oral tablet daily (recently initiated)
- Warfarin sodium 5 mg oral tablet daily
- Digoxin 0.125 mg oral tablet daily
- Verapamil hydrochloride 40 mg oral tablet three times daily
- Insulin human, isophane 70 units/mL / Regular insulin 30 units/mL (Humulin 70/30) subcutaneous injection twice daily
- Simvastatin 10 mg oral tablet daily
- Epoetin alfa 4000 units/mL injection (Epogen) 1 mL weekly
- Naproxen sodium 220 mg oral tablet as needed for arthritis pain
- Acetaminophen 325 mg oral tablet as needed for pain
- Amoxicillin 250 mg / Clavulanate 125 mg oral tablet (current course for sinusitis)
- Leucovorin 100 mg injection (recent oncology treatment)
- Oxaliplatin 5 mg/mL injection, 10 mL (recent oncology treatment)

ALLERGIES

No known drug allergies.

SOCIAL HISTORY

Ms. Echevarría is a widow who lives with her daughter and son-in-law. She immigrated from Latin America in the 1960s. She worked as a seamstress for many years before retiring in her early 70s. She has three children and seven grandchildren. Primary language is Spanish; she speaks limited English. She does not smoke or consume alcohol. She requires assistance with instrumental activities of daily living (IADLs) including medication management, financial management, and transportation. She is able to perform basic activities of daily living (ADLs) with minimal assistance, though supervision is required for safety.

FAMILY HISTORY

Mother had dementia in her late 70s, specific diagnosis unknown. Father died of myocardial infarction at age 68. One sibling with diabetes. Family history notable for cardiovascular disease and diabetes.

REVIEW OF SYSTEMS

Constitutional: No fever, chills, or weight loss. **HEENT:** Current viral and chronic sinusitis, otherwise negative. **Cardiovascular:** History of atrial fibrillation, currently rate-controlled. No chest pain or palpitations. **Respiratory:** Recent bronchitis, now resolved. No dyspnea. **Gastrointestinal:** No nausea, vomiting, or change in bowel habits. **Genitourinary:** No dysuria or incontinence. **Musculoskeletal:** Chronic knee pain from osteoarthritis. **Neurological:** As per HPI. **Psychiatric:** No depression or anxiety reported by family.

PHYSICAL EXAMINATION

Vital Signs: BP 142/78 mmHg, HR 68 bpm (irregular), RR 16/min, Temp 98.2°F, Weight 154 lbs, Height 5'2"

General: Elderly Hispanic female, well-groomed, appears stated age, no acute distress. Accompanied by family members who provide most of the history.

HEENT: Normocephalic, atraumatic. Pupils equal, round, reactive to light. Extraocular movements intact. Fundoscopic examination shows changes consistent with diabetic retinopathy. Oropharynx clear.

Cardiovascular: Irregularly irregular rhythm consistent with atrial fibrillation. No murmurs, rubs, or gallops appreciated.

Respiratory: Lungs clear to auscultation bilaterally. No wheezes, rales, or rhonchi.

Neurological Examination:

Mental Status: Alert, cooperative. Speech is fluent in Spanish with good articulation. Appears oriented to person but demonstrates confusion regarding time and place (see formal cognitive testing below).

Cranial Nerves: CN II-XII grossly intact. Visual fields full to confrontation. No facial asymmetry. Hearing intact bilaterally. Tongue midline, palate elevates symmetrically.

Motor: Normal bulk and tone throughout. Strength 5/5 in upper extremities bilaterally. Strength 4+/5 in lower extremities bilaterally, limited somewhat by arthritis. No tremor, rigidity, or bradykinesia.

Sensory: Intact to light touch and pinprick in all extremities. Proprioception intact. Vibration sense diminished distally in lower extremities, consistent with diabetic neuropathy.

Reflexes: 2+ and symmetric in biceps, triceps, and brachioradialis. 1+ at knees, 1+ at ankles. Plantar responses flexor bilaterally.

Coordination: Finger-to-nose intact bilaterally. Mild difficulty with rapid alternating movements.

Gait: Slow but steady. Mild antalgic gait due to knee osteoarthritis. No Parkinsonian features. Able to walk without assistance but uses furniture for support at times. Tandem gait not tested due to knee pain.

COGNITIVE ASSESSMENT

Mini-Mental State Examination (MMSE):

Administered in Spanish with certified interpreter assistance. Total score: 18/30

Domain	Score	Notes
Orientation to Time	1/5	Knew the season but not the date, day, month, or year
Orientation to Place	2/5	Knew the state and city but not the building, floor, or room
Registration	3/3	Successfully registered three words (ball, flag, tree)
Attention and Calculation	1/5	Serial 7s: gave only one correct subtraction
Recall	0/3	Unable to recall any of the three registered words after 5 minutes
Language	7/8	Named watch and pencil correctly. Repeated phrase. Followed 3-stage command with one error. Unable to write a complete sentence spontaneously.
Visual Construction	0/1	Unable to copy intersecting pentagons correctly (see below)

Clock Drawing Test:

The patient was asked to draw a clock showing 10 minutes past 11 o'clock. She drew a circular shape and placed some numbers around the perimeter, but number placement was disorganized with clustering on one side. She was unable to correctly place the hands to indicate the specified time. Clock drawing scored 2/10 on the standard 10-point scale, indicating significant visuospatial and executive dysfunction.

Intersecting Pentagons:

The patient attempted to copy intersecting pentagons but produced a drawing with irregular shapes that did not properly intersect. This suggests impairment in visuospatial and constructional abilities.

Delayed Recall Testing:

After a 10-minute delay, the patient was asked to recall a short story read to her in Spanish. She recalled 1 of 8 key details, demonstrating significant deficits in episodic memory and new learning capacity.

Additional Observations:

The patient demonstrated intact social pleasantries and was pleasant throughout the examination. However, she exhibited significant difficulty with tasks requiring sustained attention, working memory, and executive function. She required frequent redirection and repetition of instructions. Insight into her deficits appeared limited.

DIAGNOSTIC STUDIES

MRI Brain with and without Contrast (September 12, 2019):

Technique: Multiplanar, multisequence MRI of the brain was performed including T1-weighted, T2-weighted, FLAIR, diffusion-weighted imaging, and post-contrast T1-weighted sequences.

Findings:

- **Brain Parenchyma:** Mild generalized cerebral atrophy, greater than expected for age, with prominence of cortical sulci and ventricular enlargement. Hippocampal volumes appear reduced bilaterally, consistent with medial temporal lobe atrophy.
- **White Matter:** Moderate periventricular and deep white matter T2/FLAIR hyperintensities, consistent with chronic small vessel ischemic disease. No confluent white matter lesions.
- **Gray Matter:** No focal cortical lesions identified.
- **Posterior Fossa:** Cerebellum and brainstem appear unremarkable.
- **Vascular:** No evidence of acute or chronic territorial infarction. No hemorrhage. Major intracranial vessels demonstrate normal flow voids.
- **Extra-axial Spaces:** No subdural or epidural collections. No extra-axial masses.
- **Calvarium and Skull Base:** Unremarkable.

Impression:

1. Mild generalized cerebral atrophy with medial temporal lobe predominance, compatible with neurodegenerative process such as Alzheimer's disease.
2. Moderate chronic small vessel ischemic changes in periventricular and subcortical white matter.
3. No acute intracranial abnormality. No mass effect or midline shift.
4. No findings to suggest alternative etiology for dementia (no tumor, no normal pressure hydrocephalus, no subdural hematoma).

LABORATORY STUDIES (Reviewed from Recent Records):

- TSH: 2.4 mIU/L (normal)
- Vitamin B12: 542 pg/mL (normal)
- Complete Blood Count: Anemia present (Hgb 10.2 g/dL), consistent with known diagnosis and erythropoietin therapy
- Comprehensive Metabolic Panel: Creatinine 1.3 mg/dL (stable, consistent with CKD stage 1), glucose management variable per diabetes history
- RPR: Non-reactive
- Lipid panel: Elevated triglycerides, consistent with known hypertriglyceridemia

ASSESSMENT

1. Alzheimer's Disease, Mild-to-Moderate Stage (Primary Diagnosis)

This 92-year-old female presents with progressive cognitive decline over approximately 18-24 months, characterized by significant impairment in short-term memory, executive function, visuospatial abilities, and orientation. The clinical presentation is consistent with Alzheimer's disease based on the following:

- Insidious onset and gradual progression of cognitive symptoms
- Prominent episodic memory impairment with poor encoding and recall (0/3 on MMSE delayed recall)
- Impairment in multiple cognitive domains including executive function, visuospatial skills, and orientation
- MMSE score of 18/30, consistent with mild-to-moderate dementia
- Functional decline affecting instrumental activities of daily living, particularly medication management and financial management
- MRI findings showing generalized cerebral atrophy with medial temporal lobe predominance, supporting neurodegenerative etiology
- Exclusion of reversible causes (normal thyroid function, normal B12, no acute structural lesions)
- Positive family history of dementia in mother

The periventricular white matter changes on MRI suggest a component of cerebrovascular disease, which may be contributing to the overall cognitive picture (mixed dementia). However, the clinical presentation and pattern of deficits are most consistent with Alzheimer's disease as the primary pathology.

2. Medication Management Safety Concerns

The patient's cognitive impairment significantly compromises her ability to manage her complex medication regimen safely. This is of particular concern given:

- Polypharmacy with 12+ current medications
- Use of high-risk medications including warfarin (narrow therapeutic index, bleeding risk) and insulin (hypoglycemia risk)
- Documented episodes of medication errors (double-dosing)
- Impaired executive function and prospective memory necessary for medication adherence

3. Safety Concerns — Risk for Wandering and Getting Lost

The patient has demonstrated spatial disorientation with at least one documented episode of becoming lost in a familiar environment. This represents a significant safety risk.

4. Complex Medical Comorbidities

The patient's multiple medical conditions (atrial fibrillation, diabetes with complications, chronic kidney disease, anemia, ongoing cancer treatment) add complexity to management and may impact cognitive function. Close coordination with her primary care physician and other specialists is essential.

PLAN

1. Pharmacologic Management of Alzheimer's Disease:

a. Galantamine (Acetylcholinesterase Inhibitor):

The patient has recently been started on galantamine 4 mg daily. This is an appropriate initial dose. Galantamine is a cholinesterase inhibitor that may provide modest benefit in cognition and function in Alzheimer's disease. Given the patient's age, renal function, and complex medication regimen, we will proceed cautiously with titration:

- Continue galantamine 4 mg once daily for 4 weeks

- If tolerated without significant gastrointestinal side effects, increase to 8 mg once daily (given with food)
- After 4 weeks at 8 mg daily, consider further increase to 12 mg daily if tolerated and no concerning side effects
- Target maintenance dose: 8-12 mg daily, based on tolerability
- Monitor for common